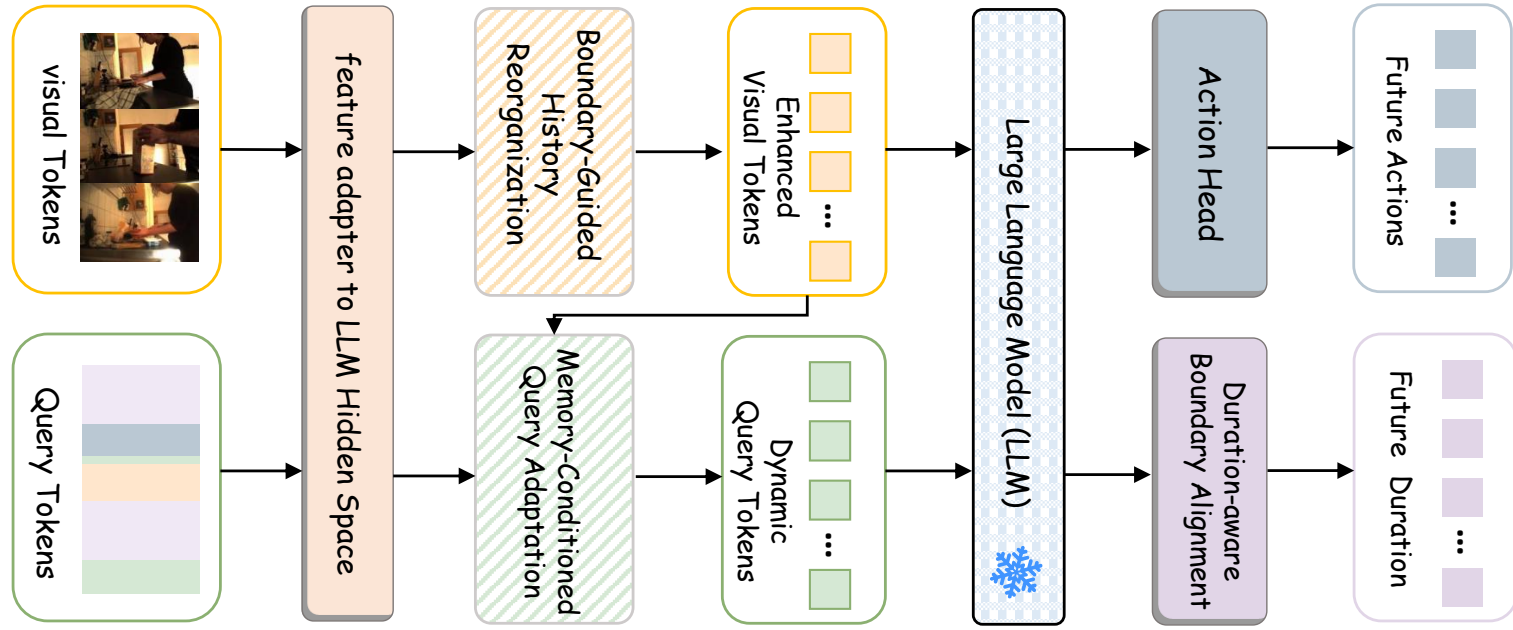
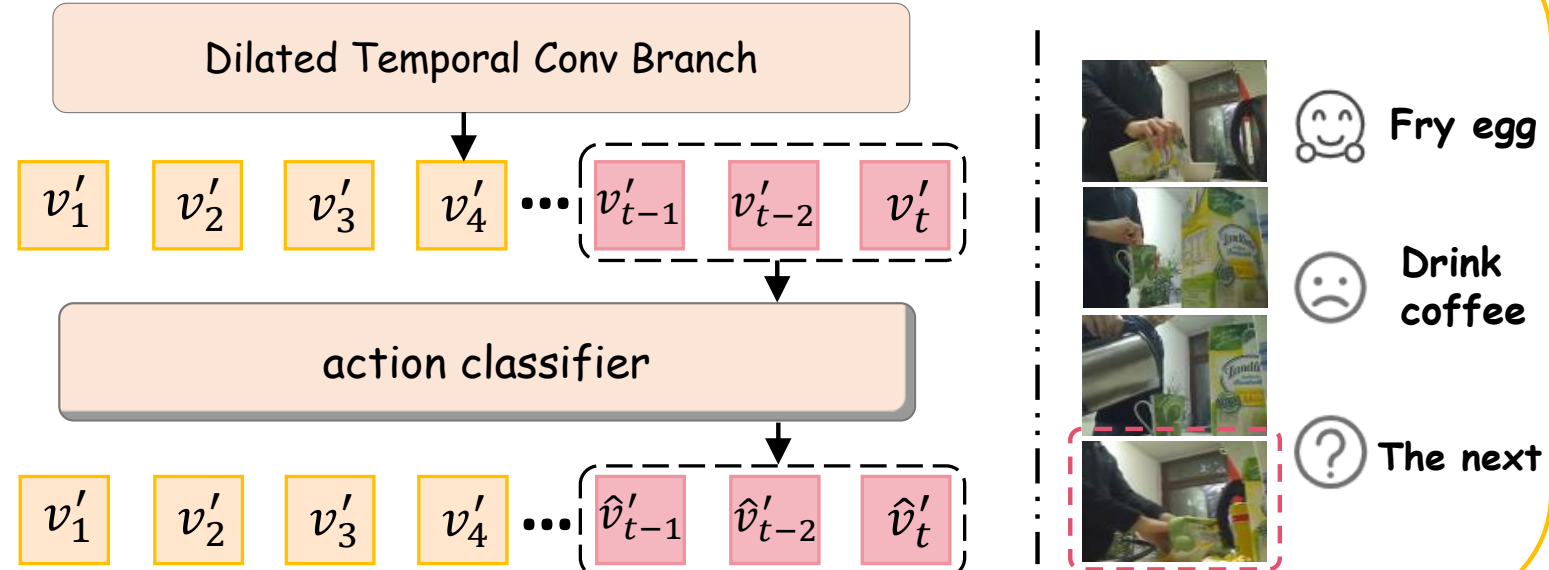




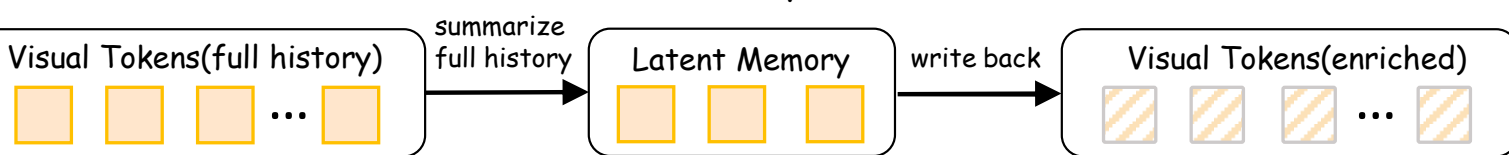
(a) ForeAct-LLM

Observed Visual Tokens $v_1, v_2, v_3, v_4, \dots, v_{t-2}, v_{t-1}, v_t$

Boundary-Proximal Memory Construction

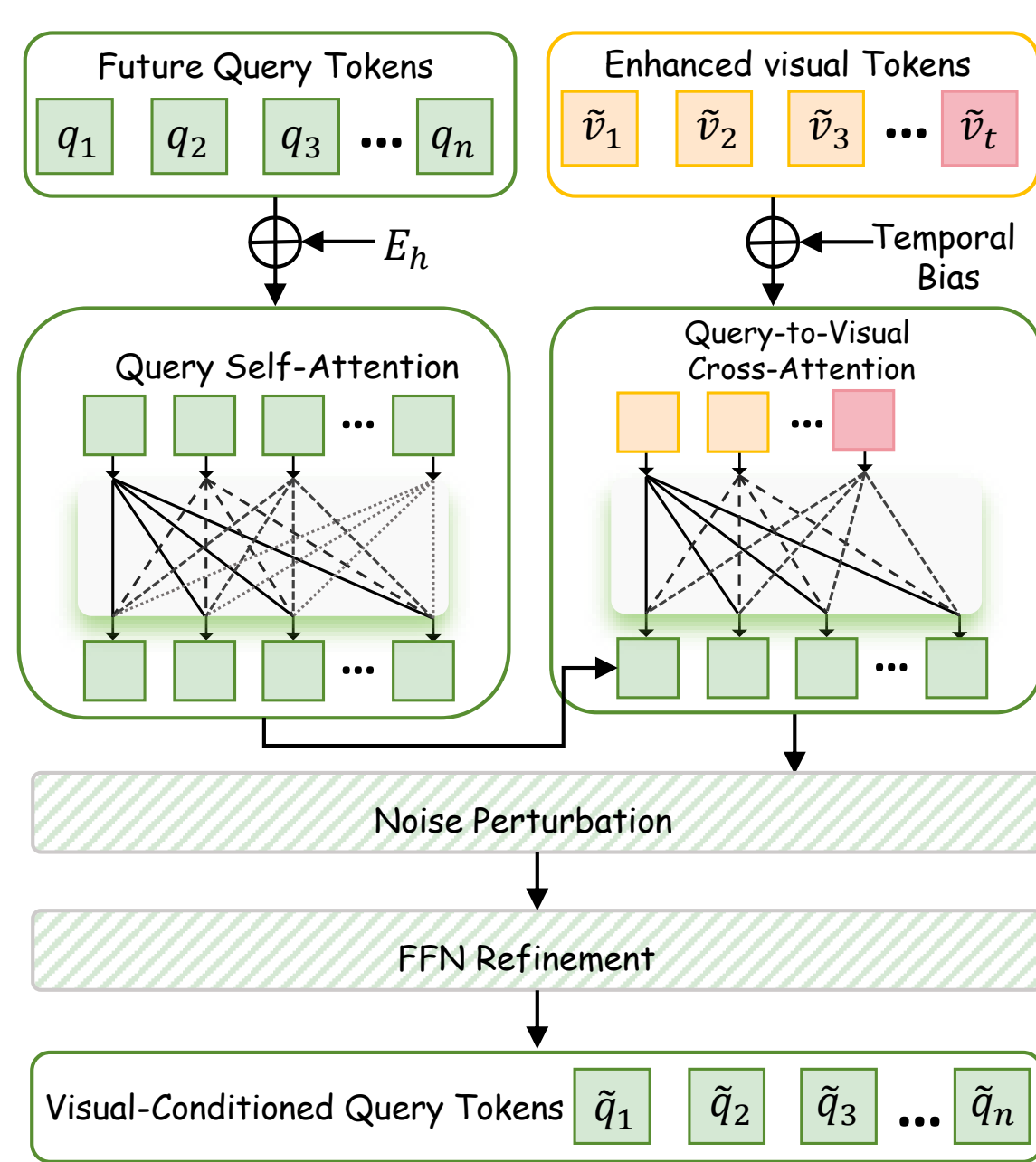


Global-Context Memory Redistribution

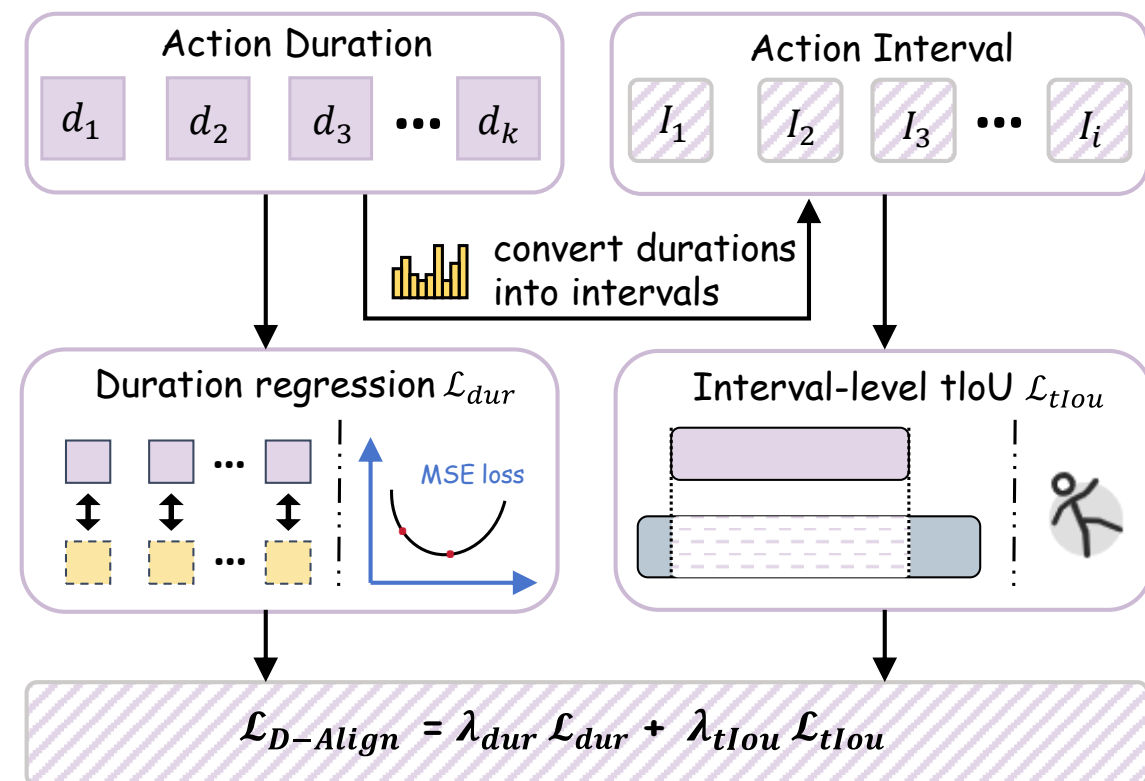


Enhanced Visual Tokens $\tilde{v}_1, \tilde{v}_2, \tilde{v}_3, \tilde{v}_4, \dots, \tilde{v}_{t-2}, \tilde{v}_{t-1}, \tilde{v}_t$

(b) Boundary-Guided History Reorganization



(c) Memory-Conditioned Query Adaptation



(d) Duration-aware Boundary Alignment